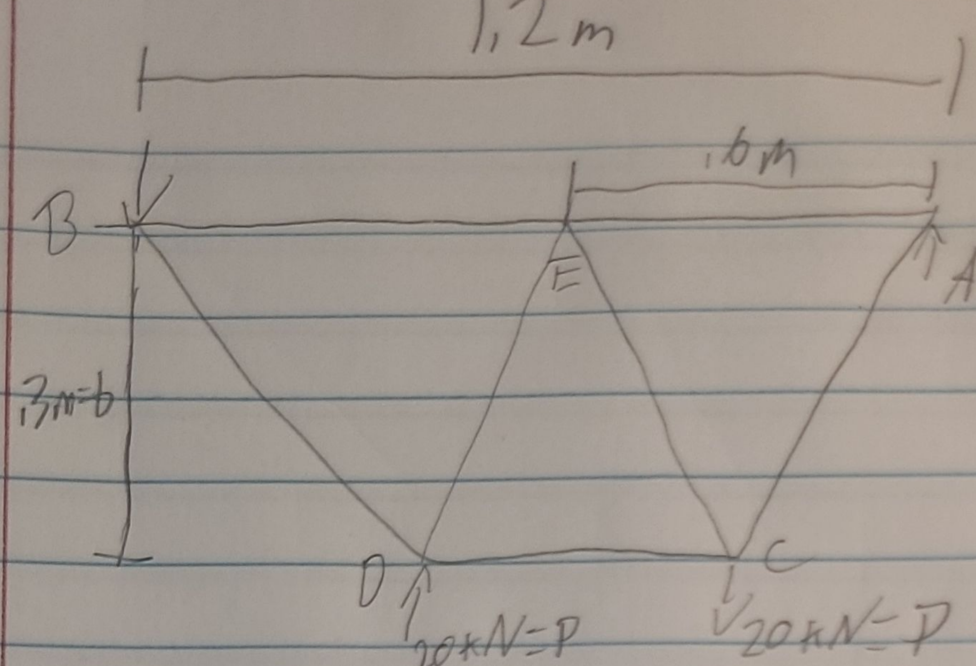

Largest internal Force: EC = 34.64 kN

Part B: $\sigma_a = \frac{\sigma_Y}{n}$ $\sigma = \frac{F}{A} \rightarrow \frac{F_{max}}{A_{min}} \leq \frac{\sigma_Y}{n}$

$F_{max} n \leq A_{min} \sigma_Y \rightarrow A_{min} = \frac{n F_{max}}{\sigma_Y}$

$A_{min} = \frac{(3.5)(34.64 \times 10^3 N)}{315 \times 10^6 Pa} = 3.849 \times 10^{-4} m^2$

$\boxed{A_{min} = 385 mm^2}$

$m = \rho V$ $V = AL_T$ $W = mg$ assuming $\rho = 7850 \, kg/m^3$

$W = \rho A L_T g$

$W = 7850(3.849 \times 10^{-4})(4.2)(9.81)$ $g = 9.81 \frac{m}{s^2}$

$W = 124.572$